

Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on understanding customer purchasing patterns using transactional data from approximately 3,900 records across multiple product categories. The objective is to identify trends in spending, segment customers effectively, analyze product preferences, and evaluate subscription behavior to support better business decision-making.

2. Dataset Summary

- **Total Records:** 3,900
- **Total Attributes:** 18

Main Data Fields Include:

- Customer details: age, gender, location, subscription status
- Purchase information: product name, category, price, season, size, color
- Behavioral aspects: discount usage, promo codes, past purchases, buying frequency, ratings, and shipping method

Missing Values:

- 37 entries were missing in the *review rating* column

3. Exploratory Data Analysis (Python)

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

The initial phase involved cleaning and preparing the dataset using Python:

- **Data Import:** Loaded the dataset using the pandas library
- **Basic Inspection:** Used functions like info() and describe() to understand structure and statistics
- **Handling Missing Values:** Filled missing review ratings with the median value of their respective product categories
- **Column Formatting:** Standardized column names into snake_case for consistency
- **Feature Creation:**
 - Grouped customers into age brackets by creating an age_group column
 - Derived a purchase_frequency_days feature from purchase-related data
- **Redundancy Check:** Found overlap between discount and promo code usage; removed the promo code column
- **Database Connection:** Uploaded the cleaned dataset into a PostgreSQL database for further querying

4. Data Analysis using SQL

Using PostgreSQL, we performed structured queries to extract business insights:

1. **Revenue Distribution by Gender** – Compared total spending between male and female customers

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Value Discount Users** – Identified customers who used discounts but still spent above average

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	60
Total rows: 839		Query complete 00:00:

3. **Top-Rated Products** – Retrieved the five products with the highest average ratings

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Impact Analysis** – Compared average order values for standard vs. express delivery

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscriber Analysis** – Evaluated spending patterns between subscribers and non-subscribers

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Driven Products** – Highlighted products most frequently purchased with discounts

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Categorized users into new, returning, and loyal groups based on purchase behavior

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Category-wise Top Products** – Identified the top three most purchased products in each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Subscription Trends in Repeat Buyers** – Analyzed whether frequent buyers tend to subscribe

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Segment** – Measured contribution of each age group to total revenue

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Power BI Dashboard

An interactive dashboard was created in Power BI to visually present the insights, making it easier to interpret trends and patterns.



6. Business Recommendations

- **Increase Subscriptions:** Offer attractive benefits to encourage more users to subscribe
- **Enhance Loyalty Programs:** Incentivize repeat customers to build a loyal user base
- **Optimize Discount Strategy:** Ensure discounts drive sales without reducing profitability
- **Promote Key Products:** Focus marketing on high-performing and top-rated products
- **Targeted Campaigns:** Direct marketing efforts toward high-value age groups and customers preferring express shipping